Payment-app project

Deploy payment app microservices application where each microservice is hosted in a Docker container, with CI/CD via Jenkins, Nexus for artifact management, SonarQube for code quality scanning, Redis for session management, and log monitoring using ELK (Elasticsearch, Logstash, Kibana) and Prometheus, and Grafana for monitoring and visualizing metrics.

Workflow:

1. Code Checkout: Fetch the latest code from the repository.
2. Code Quality Check: Analyze source code using SonarQube.
3. Frontend Build: Build the Angular project for production.
4. Backend Build: Set up Python backend with required dependencies.
5. Build Docker Images: Create Docker images for frontend and backend with the latest tags.
6. Push to Nexus: Upload built Docker images to the Nexus repository.
7. Deploy Services: Deploy frontend, backend, database, Redis, and ELK using Docker Compose.
8. Monitor Services: Set up Prometheus and Grafana for metrics visualization.(setup alert rule for alerts)
9. Logging: Forward container logs to ELK for centralized logging.
10. Caching: setup Redis for session management

Not secure 3.109.49.148:8080/job/banking%20app/

Jenkins

Search (CTRL+K) | Raghu Nandan | log out

Dashboard > banking app >

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- GitHub
- Favorite
- SonarQube
- Open Blue Ocean
- Stages
- Rename
- Pipeline Syntax

Builds

Filter /

Today

#76 7:14 PM

banking app

Edit description

Keyshell banking app

Stage View

	Declarative: Tool Install	Checkout	SonarQube Analysis	Frontend Build	Backend Build	Build Docker Images	Publish to Nexus	Deploy to EC2
Average stage times: (Average full run time: ~7min 39s)	154ms	1s	6min 14s	30s	11s	12s	14s	13s
#76 Dec 05 00:44 No Changes	168ms	1s	5min 15s	30s	9s	12s	2s	13s
#75 Dec 05 00:23 No Changes	141ms	1s	7min 13s	30s	13s	12s	25s	12s

SonarQube Quality Gate

banking-app N/A

server-side processing: In_progress

Permalinks

- Last build (#76), 7 min 57 sec ago
- Last stable build (#76), 7 min 57 sec ago
- Last successful build (#76), 7 min 57 sec ago

Not secure 65.0.85.97:8081/#browse/browse:docker-hosted:v2%2Frepository%2Fdocker-hosted%2Fpayment-app%2Ffrontend%2Ftags%2Flatest

Sonatype Nexus Repository OSS 3.73.0-12

Search components | Sign in

Browse

- Welcome
- Search
- Browse

Browse / docker-hosted

HTML View | Advanced search...

- v2
  - blobs
  - repository
    - docker-hosted
      - payment-app
        - backend
          - auth-service
            - manifests
            - tags
              - latest
          - payment-service
            - manifests
            - tags
              - latest
        - frontend
          - manifests
          - tags
            - latest
        - react-app

/v2/repository/docker-hosted/payment-app/frontend/manifests/latest

Delete asset

Summary

Repository	docker-hosted
Format	docker
Component Name	repository/docker-hosted/payment-app/frontend
Component Version	latest
Path	v2/repository/docker-hosted/payment-app/frontend/manifests/latest
Content type	application/vnd.docker.distribution.manifest.v2+json
File size	2.6 KB
Blob created	Mon Dec 02 2024 13:21:34 GMT+0530 (India Standard Time)
Blob updated	Thu Dec 05 2024 00:51:00 GMT+0530 (India Standard Time)
Last downloaded	has not been downloaded
Locally cached	true
Blob reference	default@9688eda5-e4a5-4a6b-9f6c-074f25e0cfa5
Containing repo	docker-hosted
Uploader	admin
Uploader's IP Address	3.109.49.148

Usage

Not secure 3.109.49.148:8080/blue/organizations/jenkins/banking%20app/detail/banking%20app/76/pipeline/133

banking app < 76

Pipeline | Changes | Tests | Artifacts | Logout

Branch: –
Commit: –

6m 26s
7 minutes ago

No changes
Started by user Raghu Nandan

Start → Checkout → SonarQube Analysis → Frontend Build → Backend Build → Build Docker Images → Publish to Nexus → Deploy to EC2 → End

Deploy to EC2 - 13s

Restart Deploy to EC2

node_lts — Use a tool from a predefined Tool Installation

Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.

ssh -o StrictHostKeyChecking=no -i "$SSHKEY" ${USER}@13.201.1.253 << EOF DEPLOY_DIR="/home/ubuntu/bank/payment_application" if [! -d "$DEPLOY_DIR"]; then git clone -b ${GIT_BRANCH} ${GIT_REPO} $DEPLOY_DIR else cd $DEPLOY_DIR git pull ... — Shell Script 13s

```
+ ssh -o StrictHostKeyChecking=no -i **** ubuntu@13.201.1.253
Pseudo-terminal will not be allocated because stdin is not a terminal.
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1019-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Dec  4 19:21:01 UTC 2024

  System load:  0.02               Processes:             176
  Usage of /:   65.2% of 18.33GB   Users logged in:       1
  Memory usage: 35%                IPv4 address for eth0: 172.31.37.3
  Swap usage:   0%

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

33 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
```

Payment App | Discover - Elastic | Docker and system monitoring

Not secure 13.201.1.253/dashboard

Dashboard

Make a Payment

Amount

0

Currency

USD

Description

Enter description

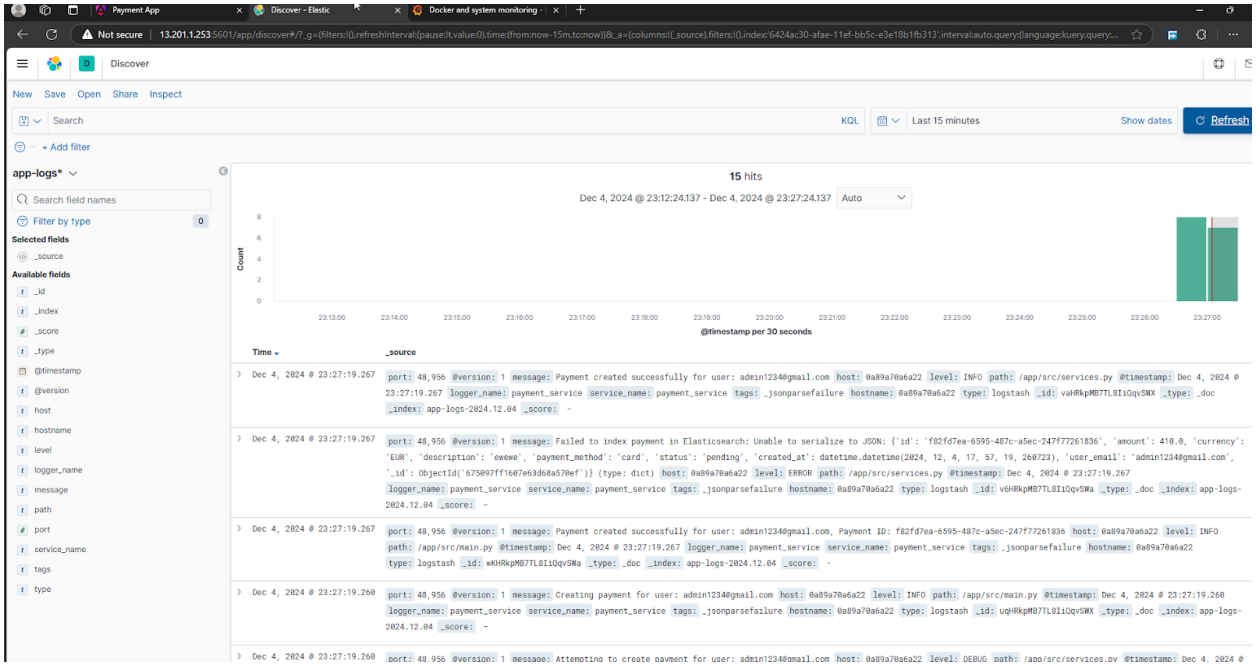
Payment Method

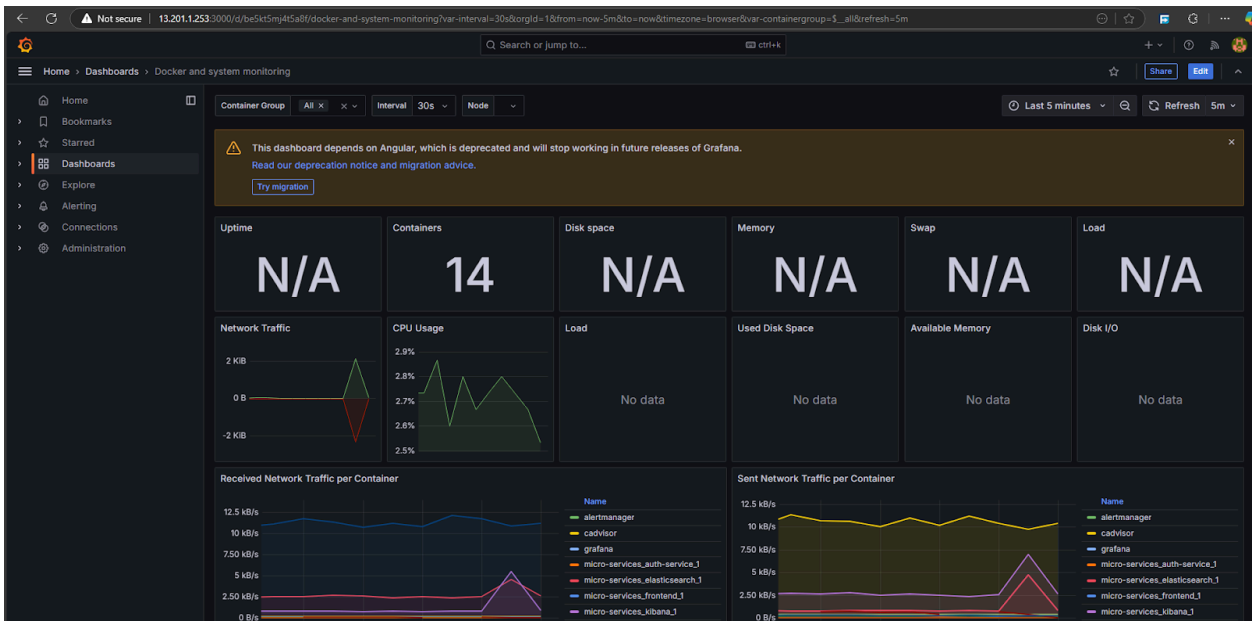
Credit Card

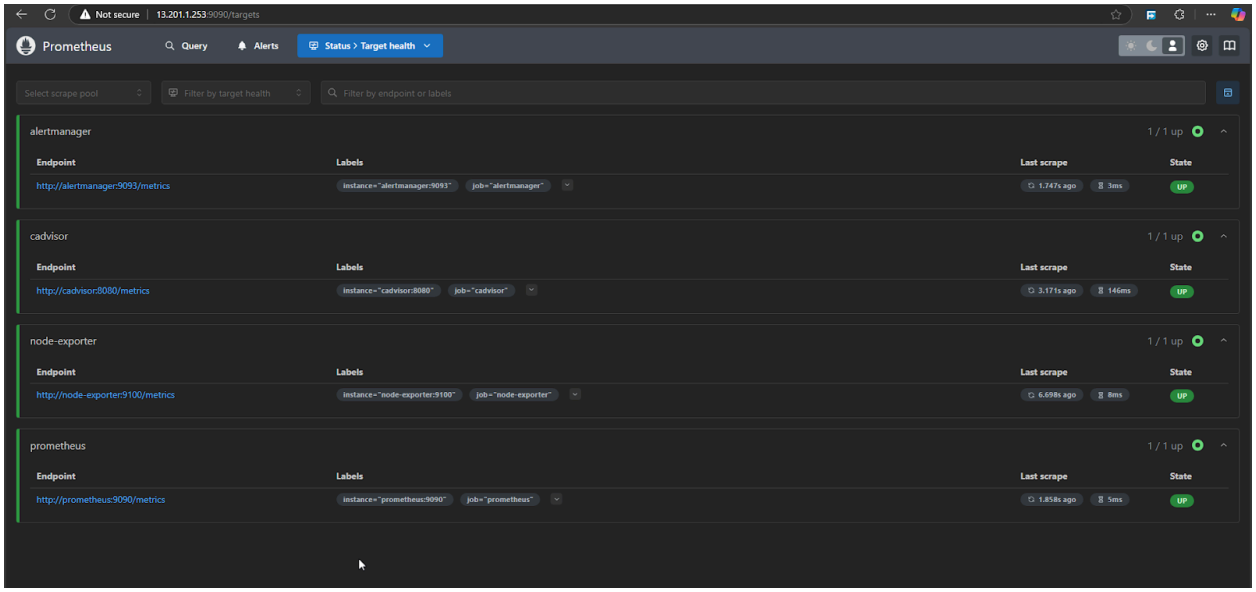
Submit Payment

Transaction History

410.00 EUR
ewewe
Dec 4, 2024, 5:57:19 PM
Pending


Payment App
Discover - Elastic
Docker and system monitoring -
Not secure | 13.201.1.253:5601/app/discover#/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-15m,to:now))&_a=(columns:!(_source),filters:!(),index:'6424ac30-afae-11ef-bb5c-e3e18b1fb313',interval:auto,query:(language:kuery,query:...
Discover
New Save Open Share Inspect
Search
KQL
Last 15 minutes
Show dates
Refresh
+ Add filter
app-logs*
Search field names
Filter by type 0
Selected fields
_source
Available fields
_id
_index
_score
_type
@timestamp
@version
host
hostname
level
logger_name
message
path
port
service_name
tags
type
15 hits
Dec 4, 2024 @ 23:12:24.137 - Dec 4, 2024 @ 23:27:24.137
Auto
Count
@timestamp per 30 seconds
Time
_source
Dec 4, 2024 @ 23:27:19.267
port: 48,956 @version: 1 message: Payment created successfully for user: admin1234@gmail.com host: 0a89a70a6a22 level: INFO path: /app/src/services.py @timestamp: Dec 4, 2024 @ 23:27:19.267 logger_name: payment_service service_name: payment_service tags: _jsonparsefailure hostname: 0a89a70a6a22 type: logstash _id: vaHRkpMB7TL8I1QqvSWX _type: _doc _index: app-logs-2024.12.04 _score: -
Dec 4, 2024 @ 23:27:19.267
port: 48,956 @version: 1 message: Failed to index payment in Elasticsearch: Unable to serialize to JSON: {'id': 'f82fd7ea-6595-487c-a5ec-247f77261836', 'amount': 410.0, 'currency': 'EUR', 'description': 'ewewe', 'payment_method': 'card', 'status': 'pending', 'created_at': datetime.datetime(2024, 12, 4, 17, 57, 19, 260723), 'user_email': 'admin1234@gmail.com', '_id': ObjectId('675097ff1607e63d68a570ef')} (type: dict) host: 0a89a70a6a22 level: ERROR path: /app/src/services.py @timestamp: Dec 4, 2024 @ 23:27:19.267 logger_name: payment_service service_name: payment_service tags: _jsonparsefailure hostname: 0a89a70a6a22 type: logstash _id: v6HRkpMB7TL8I1QqvSWa _type: _doc _index: app-logs-2024.12.04 _score: -
Dec 4, 2024 @ 23:27:19.267
port: 48,956 @version: 1 message: Payment created successfully for user: admin1234@gmail.com, Payment ID: f82fd7ea-6595-487c-a5ec-247f77261836 host: 0a89a70a6a22 level: INFO path: /app/src/main.py @timestamp: Dec 4, 2024 @ 23:27:19.267 logger_name: payment_service service_name: payment_service tags: _jsonparsefailure hostname: 0a89a70a6a22 type: logstash _id: wKHRkpMB7TL8I1QqvSWa _type: _doc _index: app-logs-2024.12.04 _score: -
Dec 4, 2024 @ 23:27:19.260
port: 48,956 @version: 1 message: Creating payment for user: admin1234@gmail.com host: 0a89a70a6a22 level: INFO path: /app/src/main.py @timestamp: Dec 4, 2024 @ 23:27:19.260 logger_name: payment_service service_name: payment_service tags: _jsonparsefailure hostname: 0a89a70a6a22 type: logstash _id: uqHRkpMB7TL8I1QqvSWX _type: _doc _index: app-logs-2024.12.04 _score: -
Dec 4, 2024 @ 23:27:19.260
port: 48,956 @version: 1 message: Attempting to create payment for user: admin1234@gmail.com host: 0a89a70a6a22 level: DEBUG path: /app/src/services.py @timestamp: Dec 4, 2024 @


Not secure | 13.201.1.253:3000/d/be5kt5mj4t5a8f/docker-and-system-monitoring?var-interval=30s&orgId=1&from=now-5m&to=now&timezone=browser&var-containergroup=$__all&refresh=5m
Search or jump to...
ctrl+k
Home > Dashboards > Docker and system monitoring
Share
Edit
Home
Bookmarks
Starred
Dashboards
Explore
Alerting
Connections
Administration
Container Group All
Interval 30s
Node
Last 5 minutes
Refresh 5m
This dashboard depends on Angular, which is deprecated and will stop working in future releases of Grafana.
Read our deprecation notice and migration advice.
Try migration
Uptime
N/A
Containers
14
Disk space
N/A
Memory
N/A
Swap
N/A
Load
N/A
Network Traffic
2 KiB
0 B
-2 KiB
CPU Usage
2.9%
2.8%
2.7%
2.6%
2.5%
Load
No data
Used Disk Space
No data
Available Memory
No data
Disk I/O
No data
Received Network Traffic per Container
12.5 kB/s
10 kB/s
7.50 kB/s
5 kB/s
2.50 kB/s
0 B/s
Name
alertmanager
cadvisor
grafana
micro-services_auth-service_1
micro-services_elasticsearch_1
micro-services_frontend_1
micro-services_kibana_1
Sent Network Traffic per Container
12.5 kB/s
10 kB/s
7.50 kB/s
5 kB/s
2.50 kB/s
0 B/s
Name
alertmanager
cadvisor
grafana
micro-services_auth-service_1
micro-services_elasticsearch_1
micro-services_frontend_1
micro-services_kibana_1




```
Setting up libglapi-mesa:amd64 (22.3.6-1+deb12u1) ...
Setting up libsctp1:amd64 (1.0.19+dfsg-2) ...
Setting up libxshmfence1:amd64 (1.3-1) ...
Setting up libbsd0:amd64 (0.11.7-2) ...
Setting up libdrm-common (2.4.114-1) ...
Setting up libkmod2:amd64 (30+20221128-1) ...
Setting up libwayland-client0:amd64 (1.21.0-1) ...
Setting up libxdmcp6:amd64 (1:1.1.2-3) ...
Setting up libxcb1:amd64 (1.15-1) ...
Setting up libxcb-xfixes0:amd64 (1.15-1) ...
Setting up libxcb-present0:amd64 (1.15-1) ...
Setting up libxcb-sync1:amd64 (1.15-1) ...
Setting up libxcb-dri2-0:amd64 (1.15-1) ...
Setting up libdrm2:amd64 (2.4.114-1+b1) ...
Setting up libxcb-randr0:amd64 (1.15-1) ...
Setting up libx11-6:amd64 (2:1.8.4-2+deb12u2) ...
Setting up libxcb-dri3-0:amd64 (1.15-1) ...
Setting up libx11-xcb1:amd64 (2:1.8.4-2+deb12u2) ...
Setting up libgbm1:amd64 (22.3.6-1+deb12u1) ...
Setting up libegl-mesa0:amd64 (22.3.6-1+deb12u1) ...
Setting up libegl1:amd64 (1.6.0-1) ...
Setting up stress-ng (0.15.06-2) ...
Processing triggers for libc-bin (2.36-9+deb12u9) ...
root@23eb2930e320:/data# stress-ng --cpu 4 --timeout 10s
stress-ng: info:  [485] setting to a 10 second run per stressor
stress-ng: info:  [485] dispatching hogs: 4 cpu
stress-ng: info:  [485] successful run completed in 10.01s
root@23eb2930e320:/data# stress-ng --cpu 10 --timeout 10s
stress-ng: info:  [490] setting to a 10 second run per stressor
stress-ng: info:  [490] dispatching hogs: 10 cpu
stress-ng: info:  [490] successful run completed in 10.02s
root@23eb2930e320:/data# stress-ng --cpu 10 --timeout 100s
stress-ng: info:  [501] setting to a 100 second (1 min, 40.00 secs) run per stressor
stress-ng: info:  [501] dispatching hogs: 10 cpu
stress-ng: info:  [501] successful run completed in 100.02s (1 min, 40.02 secs)
root@23eb2930e320:/data# stress-ng --cpu 10 --timeout 100s
stress-ng: info:  [512] setting to a 100 second (1 min, 40.00 secs) run per stressor
stress-ng: info:  [512] dispatching hogs: 10 cpu
```

Home › Alerting › Alert rules › Edit rule

Save rule | Save rule and exit | Cancel | Delete

Home
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Active notifications
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Connections
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Edit rule

1. Enter alert rule name

Enter a name to identify your alert rule.

Name: test

2. Define query and alert condition

Define query and alert condition Need help?

A | prometheus | Options 10 minutes, MD = 43200, Min. Interval = 1s | Set as alert condition

Kick start your query | Explain | Run queries | Builder | Code

Metrics browser › `rate(container_cpu_usage_seconds_total{container_name="your-container-name"}[1s]) * 100`

Options Legend: Auto Format: Time series Step: auto Type: Instant

Add query

Rule type

Select where the alert rule will be managed. Need help?

Grafana-managed | Data source-managed

The alert rule type cannot be changed for an existing rule.

Expressions

Manipulate data returned from queries with math and other operations.

B Reduce | Set as alert condition

Takes one or more time series returned from a query or an expression and turns each series into a single number.

Input: A
Function: Last
Mode: Strict

C Threshold | Alert condition

Takes one or more time series returned from a query or an expression and checks if any of the series match the threshold condition.

Input: A
IS ABOVE 1

Custom recovery threshold

Add expression | Preview

Not secure 13.201.52.222:3000/alerting/ce67cpxru8glcc/edit

Search or jump to... ctrl+k

› Edit rule

None | 10s | 20s | 30s | 40s | 50s

Pause evaluation

› Configure no data and error handling

4. Configure labels and notifications

Select who should receive a notification when an alert rule fires.

Labels

Add labels to your rule for searching, silencing, or routing to a notification policy. Need help?

No labels selected | + Add labels

Notifications

Select who should receive a notification when an alert rule fires.

Select contact point | Use notification policy

Notifications for firing alerts are routed to a selected contact point. Need help?

Alertmanager: grafana

Contact point

slack | View or create contact points

Muting, grouping and timings (optional)

5. Configure notification message

Add more context to your alert notifications. Need help?

Summary (optional)

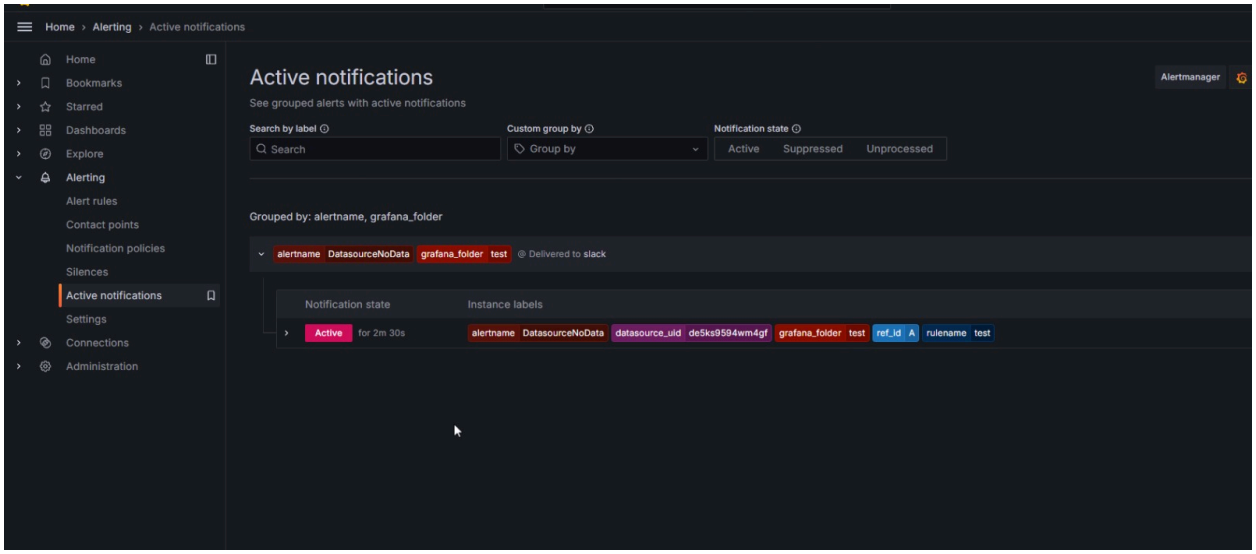
Short summary of what happened and why.

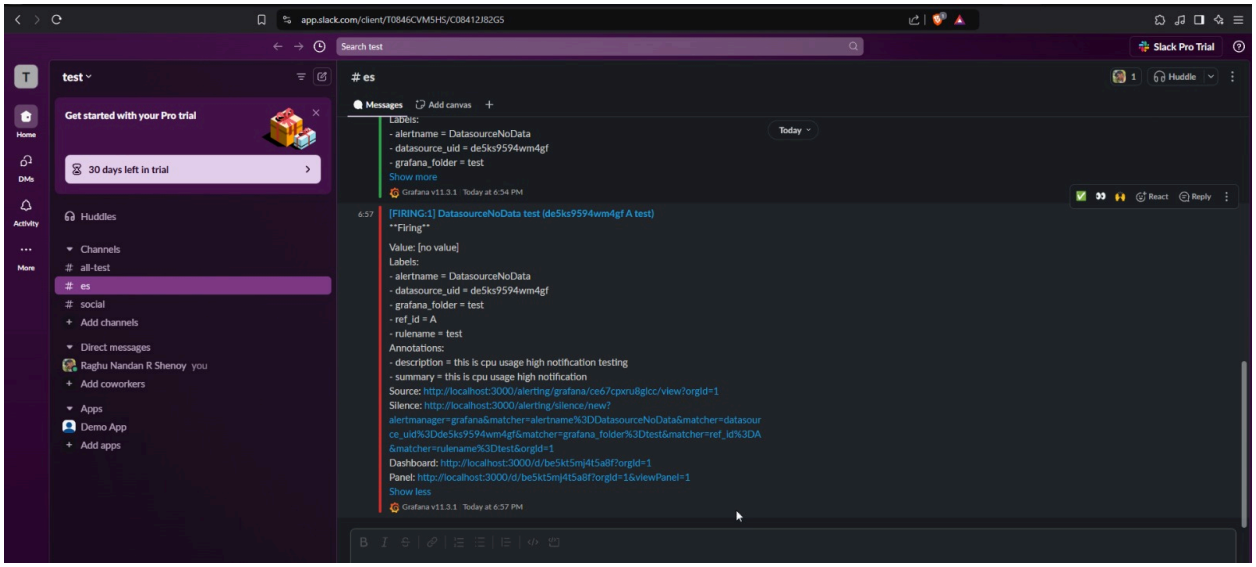
this is cpu usage high notification

Description (optional)

Description of what the alert rule does.

this is cpu usage high notification testing





Jenkins pipeline:

```
pipeline {
    agent any

    tools {
        nodejs 'node_lts'  // Make sure NodeJS is configured in Jenkins
    }

    environment {
        DOCKER_REGISTRY = '6*.*..97:8085' // Nexus Docker repo URL
```

```
        DOCKER_CREDENTIALS = credentials('nexus-credentials') // Nexus
credentials
        GIT_REPO = 'https://github.com/**********************.git'
        GIT_BRANCH = 'main'
        DEPLOY_DIR = '/home/ubuntu/bank/payment_application' // EC2
deployment directory
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', changelog: false, credentialsId:
'git_banking_cred', poll: false, url: "${GIT_REPO}"
            }
        }

        stage('SonarQube Analysis') {
            steps {
                script {
                    def scannerHome = tool 'SonarScanner' // SonarScanner
configured in Global Tools
                    withSonarQubeEnv('SonarQube') { // Configured SonarQube
environment
                        withCredentials([string(credentialsId:
'sonar_token2', variable: 'SONAR_TOKEN')]) {
                            sh "${scannerHome}/bin/sonar-scanner
-Dsonar.projectKey=banking-app -Dsonar.sources=.
-Dsonar.login=${SONAR_TOKEN}"
                        }
                    }
                }
            }
        }

        stage('Frontend Build') {
            steps {
                echo 'Building Angular frontend...'
                dir('micro-services/frontend') {
                    sh 'npm install && npm run build --prod'
                }
            }
        }
```

```
        stage('Backend Build') {
            steps {
                script {
                    def backends = ['auth_service', 'payment_service']
                    backends.each { service ->
                        dir("micro-services/backend/${service}") {
                            echo "Setting up ${service}..."
                            sh '''
                                python3 -m venv venv
                                . venv/bin/activate
                                pip install -r requirements.txt
                            '''
                        }
                    }
                }
            }
        }

        stage('Build Docker Images') {
            steps {
                echo 'Building Docker images...'
                dir('micro-services') {
                    sh '''
                    docker build -t
${DOCKER_REGISTRY}/repository/docker-hosted/payment-app/backend/auth-servic
e:latest ./backend/auth_service
                    docker build -t
${DOCKER_REGISTRY}/repository/docker-hosted/payment-app/backend/payment-ser
vice:latest ./backend/payment_service
                    docker build -t
${DOCKER_REGISTRY}/repository/docker-hosted/payment-app/frontend:latest
./frontend
                    '''
                }
            }
        }

        stage('Publish to Nexus') {
            steps {
                script {
                    withCredentials([usernamePassword(credentialsId:
'nexus-credentials',
                        usernameVariable: 'NEXUS_USER',
```

```
                        passwordVariable: 'NEXUS_PASS')]) {
                        sh "docker login -u ${NEXUS_USER} -p ${NEXUS_PASS}
${DOCKER_REGISTRY}"
                        sh "docker push
${DOCKER_REGISTRY}/repository/docker-hosted/payment-app/backend/auth-servic
e:latest"
                        sh "docker push
${DOCKER_REGISTRY}/repository/docker-hosted/payment-app/backend/payment-ser
vice:latest"
                        sh "docker push
${DOCKER_REGISTRY}/repository/docker-hosted/payment-app/frontend:latest"
                    }
                }
            }
        }

        stage('Deploy to EC2') {
            steps {
                withCredentials([sshUserPrivateKey(credentialsId:
'pk_bank',
                    keyFileVariable: 'SSHKEY',
                    usernameVariable: 'USER')]) {
                    sh '''
                        ssh -o StrictHostKeyChecking=no -i "$SSHKEY"
${USER}@13.201.1.253 << EOF
    DEPLOY_DIR="/home/ubuntu/bank/payment_application"

    if [ ! -d "$DEPLOY_DIR" ]; then
        git clone -b ${GIT_BRANCH} ${GIT_REPO} $DEPLOY_DIR
    else
        cd $DEPLOY_DIR
        git pull origin ${GIT_BRANCH}
    fi

    cat <<EOL >
$DEPLOY_DIR/micro-services/frontend/src/environments/environment.ts
export const environment = {
    production: false,
    apiUrl: 'http://<IP>:80/api',
    authServiceUrl: 'http://<IP>:80/api/auth',
    paymentServiceUrl: 'http://<IP>:80/api/payments'
};
EOL
```

```
    sed -i 's/server_name .*/server_name <IP>;/'
$DEPLOY_DIR/micro-services/nginx/conf.d/default.conf

    cd $DEPLOY_DIR/micro-services
    docker-compose down
    docker-compose up -d --build
EOF
                    '''
                }
            }
        }
    }
}
```